Vercil Juan August 20, 2026
PHYSICS 180 **Seatwork 3**

Show that the number of degenerate energy levels for an atom when summing over $2l + 1$ from $l = 0$ to $l = n - 1$, then multiplying the result by 2 is indeed $2n^2$

$$n_d = 2\sum_{l=0}^{n-1}(2l+1) = 2n^2 \tag{1}$$

Proof. We evaluate the sum:

$$\begin{aligned}
\sum_{l=0}^{n-1}(2l+1) &= 2\sum_{l=0}^{n-1} l + \sum_{l=0}^{n-1} 1 \\
&= 2\frac{(n-1)n}{2} + n \\
&= n(n-1) + n \\
&= n^2 - n + n \\
&= n^2
\end{aligned}$$

Therefore, we show (1)

$$\boxed{2\sum_{l=0}^{n-1}(2l+1) = 2n^2} \tag{2}$$

□